

Synthetic Blue-Light Exposure and Short-Term Recall

Example Research Group | Synthetic Demonstration Paper | 2026

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Abstract

This synthetic controlled experiment tests whether pre-task blue-light exposure duration changes immediate recall in a simulated adult sample.

Methods

Participants. Seventy-two synthetic adult participants were included. A seeded random allocation sequence assigned 24 participants to each of three groups.

Conditions. Participants completed 0, 30, or 60 minutes of simulated blue-light exposure before a ten-item recall task.

Analysis. The primary analysis used a one-way ANOVA across the three exposure groups. The participant was the independent unit.

Results

Immediate recall means were 8.1, 7.4, and 6.8 items for the 0-, 30-, and 60-minute groups, respectively.

The omnibus group comparison was statistically significant: **F(2,69)=5.84; p=0.004**.

Exposure duration	n	Mean recalled items
0 minutes	24	8.1
30 minutes	24	7.4
60 minutes	24	6.8

Interpretation boundary: this fictional result demonstrates extraction and verification only.

Limitations

The single-session laboratory task does not establish a persistent cognitive effect. The simulated sample, outcome, and exposure cannot be generalized to real populations.

No delayed recall, participant blinding assessment, or external replication was included.

Data and license

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